

Comments on Kratzer on attitudes

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1 Defending possible worlds semantics

In her first chapter, AK presents a broad array of techniques available to proponents of possible worlds semantics (broadly understood) to address familiar problems involving failure of intersubstitutability of modal equivalents.

The most radical idea is that we should give up on “Fregean Compositionality” (the idea that ‘the semantic value of an expression is a function of the semantic values of its parts and the way they are combined’) in order to avoid paradoxes that threaten the idea that in some environments the truth of a sentence depends on the *structured* content (tree of intensional contents) associated with some constituent.

Questions:

- The threat of paradox, as she characterizes it, seems to turn not on anything specific to language but on orthodox set theory, which rules out (thanks to the Axiom of Foundation) sets being hereditarily members of themselves. Is the proposal to give this up? And if we do, where is the threat to Fregean Compositionality?
- How are we meant to square this embrace of fine-grained content with the “modal analysis” of content nouns presented in chapter 3—an analysis which has the consequence that if P and Q are ordinary (modal-free) sentences true in the same possible worlds, every rumour [claim, suggestion] that P is also a rumour that Q?

2 ‘That’ clauses as predicates

Predicate Modification Theory: [PMT]

- ‘That P’ expresses a *property* that can be shared by things like rumors, claims, and beliefs (‘contentful entities’).
- In ‘rumor that P’, ‘rumor’ and ‘that P’ combine by *conjunction*, implemented in the H&K system by the PM (=“Predicate Modification”) compositional rule—analogue to ‘ugly rumor’ or ‘rumor which I started’ or ‘rumor in my department’ or ‘rumor according to which P’.

AK suggests that other nouns that can combine with ‘that’ clauses (e.g. ‘fact’, ‘possibility’, ‘circumstance’...) call for different readings of ‘that’ clauses.

- Why not instead say that ‘that P’ expresses a property shared by facts and possibilities as well as rumors?

For *verbs* that combine with ‘that’ clauses, AK has suggested in previous work a Davidsonian treatment where *eventualities* (like believings) can share in the property expressed by the ‘that’ clauses. This can also be extended to adjectives, on a neo-Davidsonian view where they also apply in the first instance to states (or “tropes”) rather than ordinary individuals.

3 One alternative: referential restriction.

In the chapter, AK only discusses one alternative, which I’ll call the

Referential Restriction Theory: [RRT]

- ‘that P’ refers to an entity of the sort that can instantiate the properties expressed by nouns like ‘rumor’, ‘claim’, and ‘belief’.
- In ‘rumor that P’, ‘rumor’ and ‘that P’ combine in the same way that the common noun and the name combine in constructions like ‘the coward Robert Ford’ (or maybe ‘the coward, Robert Ford’).¹

AK’s objection: rumors are things in space and time, whereas if ‘that P’ refers, it refers to something not in space and time. I’m not sure where the second half of this comes from.

A different objection that seems decisive to me: there can be many rumors [cries, proofs,...] that P (Whereas there can’t be many cowards Robert Ford).

4 An older alternative

Argument Theory: [AT]

- In constructions like ‘rumor that P’, ‘rumor’ expresses something of type $x \rightarrow e \rightarrow t$, where x is the type of ‘that P’. ‘That P’ saturates its first argument place, analogous to how ‘parent’ and ‘Alex’ have been thought to combine to make ‘parent of Alex’.²

Although AK doesn’t mention this view, she does give one argument against a different view (the ‘identity theory’ version of the PMT, see next section), which could also be turned against the AT, based on

¹AK argue that the better analogy is the comma-free one, which which she analyses as involving shifting the name into a property (*being identical to Robert Ford*) which combines with ‘coward’ by PM.

²Treating ‘of’ as vacuous as in HeimKratzerSGG. On the AT, the fact that ‘of’ isn’t required or allowed in ‘rumor that P’ is a syntactic effect that doesn’t correspond to a deep semantic difference.
<empty citation>

- (1) There is a rumor going around that you are lonely, that you haven't talked to anybody in weeks.

If the comma were replaced with 'and', this would be as unproblematic for the AT as 'I am a fan of X and of Y'. And while 'and' is normally required, we can get away without it in cases where the second part somehow amplifies the first:

- (2) I am a parent of small children, of two wonderful girls to whom I am entirely devoted.

There's also a *prima facie* strong argument (not mentioned by AK) against the AT, based on uses of 'that' clauses with 'is', like

- (3) The rumour is that he took bribes.

A proponent of the PMT could treat this just like 'The rumour is unfounded'.³

5 A problem for the PMT

An argument against the PMT (and ??) seems strong to me involves sentences like

- (4) Her remark that she had to work in the morning was a hint that it was time to leave.

Prima facie, the truth of (4) seems to require that a single thing can be *both* a remark and a hint. But if these nouns combine with 'that' clauses by PM, then the truth of (4) would entail that of

- (5) Her remark that she had to work in the morning was a remark that it was time to leave.

Avoiding this seems to require a fine-grained ontology on which no remark is *identical* to a hint, and the truth of (4) is explained by some devious reinterpretation of 'was'. But this seems hard to work out, and hard to generalize to other cases:

- (6) a. That decision is a strong indication that he's running for some office or other, and a weak indication that he's running for President.
b. That puff of smoke is a signal to the French to attack, and a signal to the Russians that any French attack will come from the west.

6 Two versions of the PMT

AK considers two versions of the PMT, which have in common the idea that each contentful entity (e.g. rumor) has a unique *content*, which is a set of possible

³I'm not sure what the AT could say about it: maybe treat it as an identity claim, with 'that he took bribes' short for 'the rumour that he took bribes'? See PottsLSPA for arguments that these 'is'-sentences are "equative" rather than "predicative".

worlds. So there is a partial function f_{content} that takes an entity x and a world w and returns the set that is its content in w , if it is contentful in w .

The two theories are then as follows:

Identity Theory: $\llbracket \text{that } P \rrbracket = \lambda x \lambda w. f_{\text{content}}(x)(w) = \llbracket P \rrbracket$.

Modal Theory: (first pass) $\llbracket \text{that } P \rrbracket = \lambda x. \lambda w. f_{\text{content}}(x)(w) \subseteq \llbracket P \rrbracket$.⁴

AK's central argument against the Identity Theory turns on this sentence

- (7) Are there any rumors circulating in my family that I live somewhere near the Czech border?

The answer could be 'yes' even if (8) is also true:

- (8) There is a rumor circulating in my family that I moved to Gmünd in Austria, and there is a rumor circulating in my family that I now live in Cieszyn in Poland, and there are exactly two rumors circulating in my family.

But on the Identity Theory, (8) entails that no rumor circulating in my family has as its exact content that I live somewhere near the Czech border, so that the answer to (7) is No. The the Modal Theory, by contrast, allows a single rumor to be both a rumor that P and a rumor that Q , even when P and Q are not equivalent.

7 Limits on monotonicity

One major worry about the Modal Theory is that it requires too *little* of a rumor for it to be a rumor that P . Bondarenko's example:

- (9) a. There is a complaint that you won an award and didn't thank anyone doesn't entail
b. There is a complaint that you won an award.

To address this problem, AK proposes that each contentful x also has at any world w , in addition to its content $f_{\text{content}}(x)(w)$, a *question addressed* and a *presupposition* that is taken for granted. We could represent these as a partition of the set of worlds compatible with the presupposition: call this $f_{\text{question}}(x)(w)$. Monotonicity is restricted by *relevance* to this question:

Modal Theory: (second pass) $\llbracket \text{that } P \rrbracket = \lambda x. \lambda w. f_{\text{content}}(x)(s) \subseteq \llbracket P \rrbracket$ and $\llbracket P \rrbracket$ is relevant to $f_{\text{question}}(x)(w)$.⁵

⁴Note: on AK's view, it's not the word 'that' itself, but rather a silent operator that's attached to P before 'that', that's responsible for transforming the meaning in this way. In some other cases, this operator is *overt*, so that there is no unit that is phonologically P and denotes a set of worlds.

⁵I'm taking liberties in writing this down: AK writes 'Given the pervasiveness of relevance considerations, I doubt that relevance requirements are imposed locally by the meanings of individual lexical items or constructions, rather than globally by default. I therefore did not include relevance

where, at least as a rough starting point:

P is relevant to $Q := P$ is incompatible with some answer to Q .

How does this address the counterexamples? As I understand it, the idea is that the rumor witnessing the truth of (9a) could have the proposition that you won an award as a *presupposition*, in which case the proposition that you won an award would not pass the relevance test, making (9b) false.

- This blocks the problem of (9a) *entailing* (9b), but I don't think it blocks the worry that the theory makes it much too easy for (9b) to be true. Wouldn't it be quite easy for a complaint to have the content *you won an award and didn't thank anyone* while *not* presupposing *you won an award*, and also addressing a question like *did you win an award and thank someone, win an award and thank no-one, or not win any award* for which *you won an award* does eliminate a cell?

8 A worry about doubts and denials

The monotonicity prediction made by the first-pass the Modal Theory is particularly implausible for things like *denials*, *doubts*, and *lies*.

- (10) a. He issued a denial that he is working for both the Russians and the Chinese.

obviously doesn't entail

- b. He issued a denial that he is working for the Russians.

And the presupposition/relevance way of limiting monotonicity seems more insufficient here.⁶

One could think that 'that' clauses mean something different when combined with "negative" content-nouns like 'denial' than when combined with "positive" ones like 'rumor'. But that will have trouble with sentences like

- (11) His insistence and her denial that they had met the previous year were both suspicious.

9 A neutral alternative

Suppose we say instead that content-bearers may have multiple "contents" (at a given world). Among these contents there may or may not be a strongest one, or a weakest one, but we give it no special semantic role, but just write:

requirements when specifying the meanings of lexical items like modal operators, and I will continue to do so.' There is an interesting general methodological attitude here to understanding what one is committed to in writing down counterexample-prone semantic generalizations like the Modal Theory (first pass).

⁶If one were going to insist that 'denial that' is monotonic, it would be more plausible to think that it's downward monotonic (if P entails Q , every denial that Q is a denial that P) than that it's upward monotonic (if P entails Q , every denial that P is a denial that Q). But both seem implausible.

Neutral Theory: $\llbracket \text{that } P \rrbracket = \lambda x. \lambda w. \llbracket P \rrbracket$ is a content of x at w .

Note that this leaves us free to agree with AK (if we want) that each *rumor* has a strongest content and has its other contents are exactly the entailments of this content that are relevant to some question, without having to say this about other kinds of content-bearers such as denials.

10 Silent and overt operators

A proponent of the PMT might conjecture that it's the word 'that' that's responsible for turning propositions into properties, e.g. for the Modal Theory (first pass):

$$\llbracket \text{that} \rrbracket = \lambda Y. \lambda x. \lambda w. f_{\text{content}}(x)(w) \subseteq Y$$

But this *isn't* AK's view. Rather, she thinks that in the examples we've been looking at, the proposition-to-property transformation is effected already by a silent operator under 'that'. And sometimes, the same job is done by an overt operator that also appear unembedded:

- (12) Das Gerücht, dass ich ein antikes Radio gestohlen haben **soll**.

Here, the logical structure is

- (13) the [rumor [that [sollen [I stole an antique radio]]]]

According to AK, the part headed by *sollen* already expresses the property of being something whose content contains only worlds where I stole an antique radio. She suggests that 'might' in English can similarly take the place of the posited covert operator. Assuming that 'might' is still analysed as contributing existential quantification over possible worlds, I think this would mean that I think the idea is that

- (14) a. There is a rumor that I might have stolen a radio.

can mean something like

- b. There is a rumor such that, at some possible world compatible with its content, I steal a radio [? and which moreover addresses a question to which the proposition that I steal a radio is relevant].

The idea that overt modals like 'sollen' and 'might' can take the place of the posited covert operator would be problematic for the Neutral Theory. That's because the relevant modals can also be used outside of 'that' clauses, and in those cases they have meanings with the characteristic logical profiles of universal/existential modal operators. AK's theory aims to unify these uses with the uses inside 'that' clauses.⁷

⁷I think the idea is that in the unembedded uses, the first argument of f_{content} is provided by the context.

However, it seems to me that the theory that ‘might’ can behave in this way makes it too easy for (14b) to be true! A rumour whose content is *compatible* with my having stolen a radio (and which is about a question to which the proposition that I stole a radio is “relevant”, such as a the question whether I am a thief) is not automatically a rumour that I might have stolen a radio.

That still leaves us with *sollen*. But here, I am worried that AK’s story will have trouble explaining how the effect arises with nouns that are not even kinda-sorta-upward-monotonic, like ‘lie’:

(15) Die Lüge, dass ich ein antikes Radio gestohlen haben soll.

I’m told this has a prominent reading where what’s a lie is that I stole an antique radio, not that I’m said to have stolen an antique radio.